

An Improvement to the Upper Bound for Marton's Covering Conjecture

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Abstract

Marton's covering conjecture studies finite sets in high-dimensional binary spaces whose pairwise sums create relatively few new elements. It predicts that every such set can be described efficiently by shifted copies of one linear subspace of comparable size. Gowers, Green, Manners, and Tao [GGMT25] proved the conjecture with exponent 12. Liao [Lia24] improved the exponent to 9. We improve it further to 8.873.

1 Introduction

For a finite subset A of an abelian group, the doubling constant $|A + A|/|A|$ measures its additive structure. Freiman-type theorems seek to describe sets with small doubling in terms of algebraic objects; over $\mathbb{F}_2^n$, the natural objects are linear subspaces and their cosets.

Ruzsa [Ruz99] recorded a sharp covering formulation in 1999 and attributed it to Marton. The conjecture asserts that whenever $|A + A| \leq K|A|$, the set A can be covered by at most $2K^C$ cosets of a single subspace whose cardinality is at most $|A|$, for an absolute constant C . The central difficulty is to obtain polynomial dependence on K , since the classical Freiman–Ruzsa argument gives a bound exponential in K .

Gowers, Green, Manners, and Tao [GGMT25] proved the covering statement with exponent $C = 12$ using an entropic descent argument. Liao [Lia24] subsequently refined the same framework in 2024, introducing a sharper admissible-penalty analysis and improving the covering exponent from 12 to 9. Green and Tao [GT09] proved the lower bound $C \geq 1.466$.

It is therefore natural to ask:

What is the optimal exponent C in Marton's covering problem?

In this work, we improve the upper bound.

1.1 Our Result

We state our result as follows.

Theorem 1.1 (Main upper bound, informal version of Theorem 4.3). *Suppose n is a positive integer, $A \subseteq \mathbb{F}_2^n$ is nonempty, and $K \geq 1$ satisfies $|A + A| \leq K|A|$. Then there is a subspace $H \leq \mathbb{F}_2^n$ with $|H| \leq |A|$ such that A is covered by at most $2K^{5+\sqrt{15}}$ translates of H .*

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Gowers, Green, Manners, and Tao [GGMT25] obtained the covering exponent 12. Liao [Lia24, Theorem 2] improved it to 9. We further improve it to 8.873. For the optimal exponent C_* in Marton’s covering problem, the best known lower bound is $C_* \geq 1.466$, proved by Green and Tao [GT09].

2 Technique Overview

Section 2.1 reviews the Gowers–Green–Manners–Tao descent, including the four-copy fibring setup, the primary-candidate constraints, and the characteristic-two endgame. It then derives the closing inequality in Eq. (4) and explains how it leads to covering exponent 12.

Section 2.2 identifies the distance estimates and descent ingredients that Liao reuses unchanged from GGMT, and isolates his new admissible-penalty aggregation. It derives Liao’s improved inequality in Eq. (7) and records the Kullback–Leibler dense-slice conversion that gives covering exponent 9.

Section 2.3 states precisely which ingredients we retain from GGMT and Liao and isolates our only new step: summing Liao’s three fibre-side bounds without averaging them with the sum-side bounds. This yields Eq. (10) and the covering exponent $5 + \sqrt{15} < 8.873$.

2.1 Summary of GGMT25

The shared descent setup. For G -valued random variables X and Y , with independent copies X' and Y' , define their entropic Ruzsa distance by $d[X; Y] := H[X' - Y'] - \frac{1}{2}H[X] - \frac{1}{2}H[Y]$. It is nonnegative and vanishes precisely when X and Y are uniform on cosets of one common subgroup [Tao10]. Consequently, if U_A is uniform on a set A of doubling at most K , then $d[U_A; U_A] \leq \log K$.

Take four independent variables X_1, X_2 distributed as X and Y_1, Y_2 distributed as Y . Set $T := X_1 + Y_1$, $\bar{T} := X_2 + Y_2$, $W := X_1 + X_2$, and $\bar{W} := Y_1 + Y_2$. Also set $S := T + \bar{T}$, $V := X_1 + Y_2$, and $d := d[X; Y]$. These variables satisfy the two fibring identities of GGMT [GGMT25, Corollary 4.2], which express each of $2d - I_1$ and $2d - I_2$ as the sum of a sum-side distance and a fibre-side distance, where $I_1 = I[T : V \mid S]$ and $I_2 = I[T : W \mid S]$. Testing the four resulting pairs against the descent potential gives, whenever none of them decreases the potential,

$$u := 2\eta d - I_1 \geq 0, \quad I_2 - 2\eta d \leq \frac{\eta}{1 - \eta} u. \quad (1)$$

The constraints in Eq. (1), the characteristic-two identity $T + V + W = 0$, and the entropic Balog–Szemerédi–Gowers estimate of Gowers, Green, Manners, and Tao [GGMT25, Lemma A.2] are the common inputs to all three arguments.

How GGMT obtain Eq. (4). GGMT use penalties given by Ruzsa distance to two fixed reference variables and minimise distance plus η times the total penalty. For the three conditioned endgame pairs coming from (T, V, W) , let $\tilde{\delta}$ be the sum of their three mutual informations, and let P_G be the total increase of the six reference-distance penalties.

We have

$$\tilde{\delta} \leq 6\eta d - \frac{1 - 5\eta}{1 - \eta} u, \quad P_G \leq (6 - 3\eta)d + 3u. \quad (2)$$

The first bound in Eq. (2) follows from the first estimate in Section 7 of [GGMT25]. The second bound follows from the second estimate there.

We have

$$d \leq (1 + \frac{\eta}{3})\tilde{d} + \frac{\eta}{3}P_G. \quad (3)$$

where the step follows from the symmetrised endgame lemma of GGMT.

We have

$$d \leq (8\eta + \eta^2)d - [(1 + \frac{\eta}{3})\frac{1 - 5\eta}{1 - \eta} - \eta]u \leq (8\eta + \eta^2)d. \quad (4)$$

The first step follows from Eqs. (2) and (3), and simple algebra. The second step follows from Eq. (1), since $u \geq 0$ and the coefficient of u in the first line is nonpositive in the relevant range. For $d > 0$ this is contradictory when $8\eta + \eta^2 < 1$. GGMT take $\eta = 1/9$; their entropic coefficient is then $\eta^{-1} + 2 = 11$, and the standard conversion from the entropic statement to a covering statement costs one further power of K , giving exponent 12.

2.2 Summary of Liao24

What Liao reuses from GGMT. Liao keeps unchanged the entropic Ruzsa distance and its equality case, the compactness/minimisation descent, the four independent copies and the sums $T, \bar{T}, V, W, \bar{W}, S$, the two fibring identities [Lia24, Eqs. (1) and (2)], the same four primary candidates, and therefore the same two constraints on u, I_1, I_2 in Eq. (1). He also keeps the identity $T + V + W = 0$, the same three conditioned endgame candidates, and the GGMT entropic Balog–Szemerédi–Gowers bound [Lia24, Eq. (4)], which bounds the sum of their three distances by $3I_1 + 6I_2$. Thus Liao does not change the distance side of the descent.

What Liao changes. Liao abstracts the reference-distance penalties into two admissible functions τ_A, τ_B and uses

$$\phi[X; Y] := d + \eta\tau_0, \quad \tau_0 := \tau_A(X) + \tau_B(Y). \quad (5)$$

Admissibility means continuity and translation invariance together with the two entropy-controlled rules for adding an independent variable and conditioning on a correlated one [Lia24, Lemma 7]. For each of the three endgame candidates, Liao proves a sum-side penalty bound [Lia24, Eqs. (12)–(14)] and a fibre-side penalty bound [Lia24, Eqs. (15)–(17)]. He averages the two bounds for each candidate and then sums the three averages. This is Liao’s Eq. (18):

$$\tau_{\text{eg}} \leq 3\tau_0 + 6d + (I_2 - I_1). \quad (6)$$

Figure 1 illustrates Eq. (6). If none of the three endgame candidates decreases ϕ , their total potential is at least $3\phi[X; Y] = 3d + 3\eta\tau_0$.

We have

$$d \leq I_1 + 2I_2 + \frac{\eta}{3}(\tau_{\text{eg}} - 3\tau_0) \leq I_1 + 2I_2 + \frac{\eta}{3}(6d + I_2 - I_1) \leq 8\eta d + \frac{-3 + 10\eta}{3(1 - \eta)}u \leq 8\eta d. \quad (7)$$

The first step follows from the absence of descent, Eq. (5), and the GGMT distance bound [Lia24, Eq. (4)], followed by simple algebra. The second step follows from Eq. (6). The third step follows from Eq. (1) and simple algebra. The last step follows because $(-3 + 10\eta)/(3(1 - \eta)) < 0$ for $\eta < 1/8$ and $u \geq 0$ by Eq. (1). This is the precise endgame improvement: Liao removes the $\eta^2 d$ loss from GGMT’s Eq. (4). Hence Eq. (7) is impossible for $d > 0$ and $\eta < 1/8$.

For the covering statement, Liao makes one further change that is separate from the endgame calculation. He takes $\tau_A = \tau_B = \frac{1}{2}(\tau^- + \tau^+)$, where τ^- is the least Kullback–Leibler divergence to a smoothing of U_A and $\tau^+(Z) = \tau^-(Z) + H[Z] - H[U_A]$. Its exact dense-coset identity [Lia24, Lemma 11 and Claim 12] converts the coefficient 8 into a coset of relative density K^{-8} . Ruzsa covering costs one more power of K , giving exponent 9.

2.3 Summary of Our Upper Bound

The common descent framework From GGMT we reuse the entropic Ruzsa calculus and equality case, the minimisation/descent framework, the four-copy construction and fibring identities, the four primary candidates and the constraints in Eq. (1), the identity $T + V + W = 0$, and the entropic Balog–Szemerédi–Gowers distance bound. From Liao we reuse the admissible-penalty potential ϕ , his three conditioned endgame candidates, his Kullback–Leibler penalty and exact dense-slice conversion, and in particular his three fibre-side estimates [Lia24, Eqs. (15)–(17)]. None of these ingredients is modified.

Direct aggregation of the fibre-side estimates We change only the aggregation of the three endgame penalties. Liao’s aggregation [Lia24, Eq. (18)] averages the sum-side estimates [Lia24, Eqs. (12)–(14)] with the fibre-side estimates [Lia24, Eqs. (15)–(17)]. We discard the sum-side estimates and sum the three fibre-side estimates directly. Translation invariance identifies the first conditioned pair because $V = T + \bar{W}$, while exchanging Y_1 and Y_2 identifies the other two. Consequently,

$$\tau_{\text{eg}} \leq \tau_{\bar{W}} + e + 2(\tau_{\bar{T}} + b). \quad (8)$$

Figure 2 illustrates Eq. (8).

We have

$$\tau_{\text{eg}} \leq 3\tau_0 + (6 + 3\eta)d - \frac{5 + \eta}{2}I_1 + \frac{\eta - 1}{2}I_2. \quad (9)$$

where the step follows from Eq. (8) by applying the same admissibility bounds and primary-candidate constraints as Liao. The distance part is still $3I_1 + 6I_2$.

We have

$$\begin{aligned} d &\leq I_1 + 2I_2 + \frac{\eta}{3}(\tau_{\text{eg}} - 3\tau_0) \\ &\leq I_1 + 2I_2 + \frac{\eta}{3}[(6 + 3\eta)d - \frac{5 + \eta}{2}I_1 + \frac{\eta - 1}{2}I_2] \\ &\leq (8\eta - \eta^2)d + \frac{-6 + 23\eta - 5\eta^2}{6(1 - \eta)}u \\ &\leq (8\eta - \eta^2)d. \end{aligned} \quad (10)$$

The first step follows from the absence of descent, Eq. (5), and the GGMT distance bound [Lia24, Eq. (4)], followed by simple algebra. The second step follows from Eq. (9). The third step follows from Eq. (1) and simple algebra. The last step follows because the coefficient of u is negative for $\eta < 1/7$ and $u \geq 0$ by Eq. (1). Thus we retain all of Liao’s machinery but recover the term $\eta^2 d$ that his averaging loses. For $d > 0$, Eq. (10) is contradictory when $8\eta - \eta^2 < 1$. Its sharp relevant threshold is $\eta < 4 - \sqrt{15}$, so the penalty coefficient is $1/(4 - \sqrt{15}) = 4 + \sqrt{15}$. Liao’s dense-slice conversion is unchanged, and Ruzsa covering again adds one power of K . The resulting covering exponent is $5 + \sqrt{15} < 8.873$. The complete proof is given in Section 4.

3 Preliminaries

Section 3.1 defines the notation for binary vector spaces, entropy, and conditional information quantities. Section 3.2 formulates Marton’s covering problem and introduces its optimal exponent. Section 3.3 introduces the entropic Ruzsa distance and admissible penalties and records the dense-slice covering tool used in the final argument.

Liao: average both bounds for each penalty block

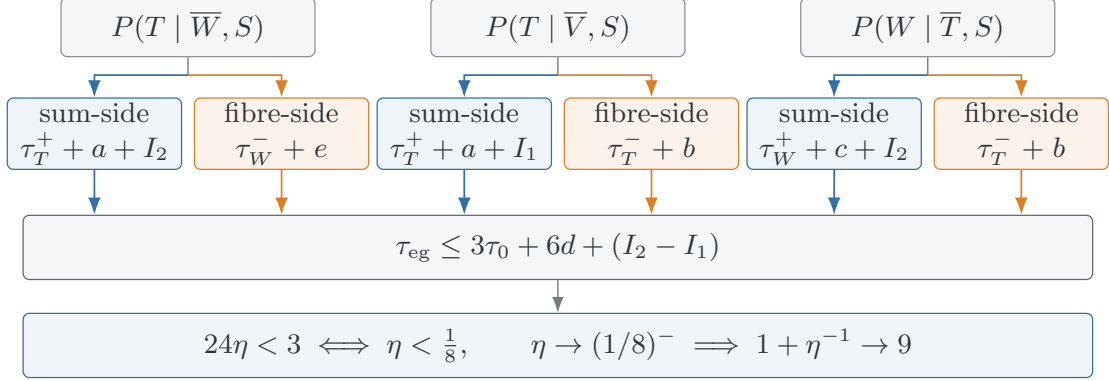


Figure 1: Liao’s endgame aggregation [Lia24]. The GGMT distance estimate is unchanged. On the penalty side, Liao averages a sum-side and a fibre-side bound for each of the three conditioned candidates, then sums the three averages to obtain Eq. (6).

Ours: retain only the fibre-side bounds

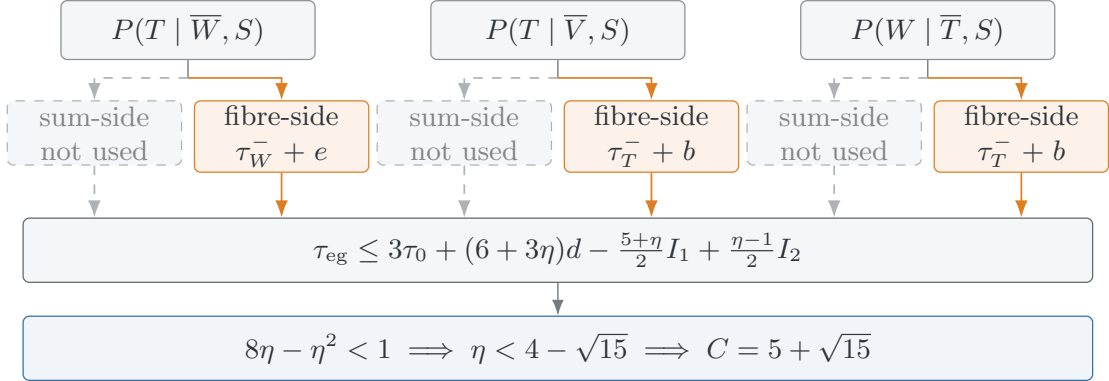


Figure 2: Our fibre-only aggregation. All distance estimates and primary-candidate constraints are reused. The only change is that the three fibre-side penalty bounds are retained and summed directly, rather than averaged with the sum-side bounds.

3.1 Notation

All logarithms have base 2. For a positive integer n we write $G = \mathbb{F}_2^n$, and we write $H \leq G$ to mean that H is a subspace of G . For $A \subseteq G$ we write $A + A := \{a + a' : a, a' \in A\}$, and since G has characteristic 2 subtraction and addition agree. For $x \in G$ the set $x + H$ is a coset of H , also called a translate of H . For a G -valued random variable X , let $H[X]$ denote its Shannon entropy, and let U_S denote the uniform distribution on a finite nonempty set $S \subseteq G$. For jointly distributed random variables, write

$$H[X | Z] := \mathbb{E}_{z \sim Z}[H[X | Z = z]]$$

and

$$I[X : Y | Z] := H[X | Z] + H[Y | Z] - H[X, Y | Z]$$

for conditional entropy and conditional mutual information. For probability distributions P and Q on G , write

$$D_{\text{KL}}(P \parallel Q) := \sum_{x: P(x) > 0} P(x) \log \frac{P(x)}{Q(x)},$$

with value $+\infty$ if $P(x) > 0$ and $Q(x) = 0$ for some x . For a function τ on distributions, we write

$$\tau(X | Z) := \mathbb{E}_{z \sim Z}[\tau(X | Z = z)].$$

3.2 The covering problem

We quantify Marton's covering conjecture through the exponent C governing the polynomial dependence on the doubling constant K . The optimal value C_* is the infimum formalized below.

Definition 3.1 (Marton's polynomial Freiman–Ruzsa problem [Ruz99]). *Determine the infimum of all real numbers $C \geq 0$ for which the following assertion holds. Suppose n is a positive integer, $A \subseteq \mathbb{F}_2^n$ is nonempty, and $K \geq 1$ satisfies $|A + A| \leq K|A|$. Then A is covered by at most $2K^C$ translates of a subspace $H \leq \mathbb{F}_2^n$ with $|H| \leq |A|$.*

3.3 Entropic Ruzsa distance and penalties

We first recall the entropic analogue of Ruzsa distance.

Definition 3.2 (Entropic Ruzsa distance [Tao10]). *Suppose X and Y are G -valued random variables. Let X' and Y' be independent copies of X and Y . The entropic Ruzsa distance is*

$$d[X; Y] := H[X' - Y'] - \frac{1}{2}H[X] - \frac{1}{2}H[Y].$$

We also use two conditional forms.

Definition 3.3 (Jointly conditioned entropic Ruzsa distance). *Suppose X, Y, Z are jointly distributed. The jointly conditioned entropic Ruzsa distance is*

$$d[(X; Y) | Z] := \mathbb{E}_{z \sim Z}[d[X | Z = z; Y | Z = z]],$$

If the conditioning variable is a tuple, its components are sampled jointly.

Definition 3.4 (Independently conditioned entropic Ruzsa distance). *Suppose (X, Z) is independent of (Y, W) . The independently conditioned entropic Ruzsa distance is*

$$d[X | Z; Y | W] := \mathbb{E}_{\substack{z \sim Z \\ w \sim W}} [d[X | Z = z; Y | W = w]],$$

where z and w are sampled independently from their marginals.

Liao's abstract penalty lemma [Lia24, Lemma 7] uses the following notion of admissibility.

Definition 3.5 (Admissible penalty). *A continuous real-valued function τ on G -valued distributions is admissible if the following properties hold. Suppose X and Y are independent G -valued random variables. Suppose Z is jointly distributed with X . Then*

$$\begin{aligned} \tau(X + Y) &\leq \tau(X) + \frac{1}{2}(\mathbb{H}[X + Y] - \mathbb{H}[X]), \\ \tau(X | Z) &\leq \tau(X) + \frac{1}{2}(\mathbb{H}[X] - \mathbb{H}[X | Z]), \\ \tau(X + s) &= \tau(X) \quad \text{for every } s \in G. \end{aligned}$$

We state the following dense-slice-to-covering lemma of Liao [Lia24, Lemma 9], which isolates the covering argument from Gowers, Green, Manners, and Tao [GGMT25, Appendix B].

Lemma 3.6 (Dense slice to covering, [Lia24, Lemma 9]). *Suppose $A \subseteq G$ is nonempty. Suppose $|A + A| \leq K|A|$. Suppose $R \geq 1$. Suppose $V \leq G$ and $t \in G$ satisfy*

$$|A \cap (V + t)| \geq \frac{\max\{|A|, |V|\}}{R}.$$

Then A is covered by at most $2KR$ translates of a subspace $V' \leq G$ with $|V'| \leq |A|$.

4 Proof of the Upper Bound

The goal of this section is to prove Theorem 4.3. The argument begins with Liao's abstract penalty lemma, which gives coefficient 8 under the admissibility hypotheses of Definition 3.5 [Lia24, Lemma 7] and recasts the descent argument of Gowers, Green, Manners, and Tao in admissible-penalty form [GGMT25, Proposition 2.1 and the proof of Theorem 1.8].

The proof of the next lemma follows Liao's descent argument [Lia24, Lemma 8]. We retain the four primary candidates and the three endgame candidates, and modify the final aggregation by combining Liao's fibre-side inequalities [Lia24, Eqs. (15)–(17)] directly with the bounds forced by failure of the primary candidates.

Lemma 4.1 (Improved penalty lemma). *Suppose τ_A and τ_B are admissible penalties on $G = \mathbb{F}_2^n$. Suppose X and Y are G -valued random variables. Then there exists a subspace $V \leq G$ such that*

$$\tau_A(U_V) + \tau_B(U_V) \leq \tau_A(X) + \tau_B(Y) + (4 + \sqrt{15})d[X; Y].$$

Proof. Fix a parameter.

$$0 < \eta < 4 - \sqrt{15} < \frac{1}{7} < 1. \tag{11}$$

The first two inequalities specify the parameter range. The third and last inequalities follow from simple algebra.

Set

$$\phi[X; Y] := d[X; Y] + \eta(\tau_A(X) + \tau_B(Y)). \quad (12)$$

We show that every pair with positive Ruzsa distance admits a strict decrease of ϕ .

Let X_1, X_2 be independent copies of X . Let Y_1, Y_2 be independent copies of Y . Suppose all four variables are independent. Define

$$\begin{aligned} T &:= X_1 + Y_1, & \bar{T} &:= X_2 + Y_2, \\ V_0 &:= X_1 + Y_2, & \bar{V} &:= X_2 + Y_1, \\ W &:= X_1 + X_2, & \bar{W} &:= Y_1 + Y_2, \\ S &:= X_1 + X_2 + Y_1 + Y_2. \end{aligned} \quad (13)$$

The variables in Eq. (13) satisfy

$$\begin{aligned} S &= T + \bar{T} = V_0 + \bar{V} = W + \bar{W}, \\ V_0 &= T + \bar{W} = T + W + S = W + \bar{T}, & W + S &= \bar{W}. \end{aligned} \quad (14)$$

All identities follow from Eq. (13) and the characteristic-two identity $x + x = 0$.

Set

$$\begin{aligned} I_1 &:= I[T : V_0 \mid S], \\ I_2 &:= I[T : W \mid S] = I[V_0 : W \mid S]. \end{aligned} \quad (15)$$

The second equality in Eq. (15) follows from the symmetry obtained by exchanging the i.i.d. variables Y_1 and Y_2 , which interchanges T and V_0 while fixing W and S . Write

$$\begin{aligned} d &:= d[X; Y], & \tau_0 &:= \tau_A(X) + \tau_B(Y), \\ a &:= d[T; \bar{T}], & b &:= d[X_1 \mid T; Y_2 \mid \bar{T}], \\ c &:= d[W; \bar{W}], & e &:= d[X_1 \mid W; Y_1 \mid \bar{W}]. \end{aligned} \quad (16)$$

The terms b and e use the independently conditioned Ruzsa distance in Definition 3.4.

We have

$$a + b = 2d - I_1, \quad c + e = 2d - I_2. \quad (17)$$

where both identities follow from the four-variable fibring identity of Gowers, Green, Manners, and Tao [GGMT25, Corollary 4.2], written in Liao's notation [Lia24, Eqs. (1) and (2)].

We also need the third fibring identity.

We have

$$\begin{aligned} d[X; X] + d[Y; Y] &= a + d[X_1 \mid T; X_2 \mid \bar{T}] + I_2 \\ &= a + d[X_1 \mid T; Y_2 \mid \bar{T}] + I_2 \\ &= a + b + I_2 \\ &= a + (2d - I_1 - a) + I_2 \\ &= 2d + I_2 - I_1. \end{aligned} \quad (18)$$

The first step follows from the four-variable fibring identity of Gowers, Green, Manners, and Tao [GGMT25, Corollary 4.2]. The second step follows because, after conditioning on $\bar{T}$, the variables X_2 and Y_2 differ by the fixed translation $\bar{T}$, so translation invariance applies. The third step follows from Eq. (16). The fourth step follows from Eq. (17). The last step follows from simple algebra. This is Liao's third fibring identity [Lia24, Eq. (3)].

Define the four primary penalty totals.

$$\begin{aligned}
\tau_T^+ &:= \tau_A(T) + \tau_B(\bar{T}), \\
\tau_T^- &:= \tau_A(X_1 | T) + \tau_B(Y_2 | \bar{T}), \\
\tau_W^+ &:= \tau_A(W) + \tau_B(\bar{W}), \\
\tau_W^- &:= \tau_A(X_1 | W) + \tau_B(Y_1 | \bar{W}).
\end{aligned} \tag{19}$$

We have

$$\tau_T^+, \tau_T^- \leq \tau_0 + d, \tag{20}$$

$$\tau_W^+, \tau_W^- \leq \tau_0 + \frac{1}{2}d[X; X] + \frac{1}{2}d[Y; Y] = \tau_0 + \frac{1}{2}(2d + I_2 - I_1) = \tau_0 + d + \frac{1}{2}(I_2 - I_1). \tag{21}$$

The step in Eq. (20) follows from admissibility. The first step in Eq. (21) follows from admissibility. The second step follows from Eq. (18). The last step follows from simple algebra.

These are precisely Liao's inequalities [Lia24, Eqs. (5)–(8)].

The four primary candidates are the two sum pairs $(T, \bar{T})$ and $(W, \bar{W})$ and the two fibre pairs $(X_1 | T, Y_2 | \bar{T})$ and $(X_1 | W, Y_1 | \bar{W})$. The potential of a fibre pair is averaged over independent conditioning values. Thus a strict decrease for an averaged fibre potential gives a strict decrease after fixing those values. If any primary candidate strictly decreases ϕ , then we are done. Suppose that no primary candidate decreases ϕ .

We have

$$a + \eta\tau_T^+ \geq d + \eta\tau_0, \tag{22}$$

$$c + \eta\tau_W^+ \geq d + \eta\tau_0, \tag{23}$$

where all four steps follow from the assumption that none of the four primary candidates decreases the potential in Eq. (12).

We prove the bound for b .

We have

$$b = 2d - I_1 - a \leq 2d - I_1 - (d - \eta(\tau_T^+ - \tau_0)) \leq 2d - I_1 - (1 - \eta)d = (1 + \eta)d - I_1. \tag{24}$$

The first step follows from Eq. (17). The second step follows from Eq. (22). The third step follows from Eq. (20). The last step follows from simple algebra.

We prove the bound for e .

We have

$$\begin{aligned}
e &= 2d - I_2 - c \\
&\leq 2d - I_2 - (d - \eta(\tau_W^+ - \tau_0)) \\
&\leq 2d - I_2 - (1 - \eta)d + \frac{\eta}{2}(I_2 - I_1) \\
&= (1 + \eta)d - I_2 + \frac{\eta}{2}(I_2 - I_1).
\end{aligned} \tag{25}$$

The first step follows from Eq. (17). The second step follows from Eq. (23). The third step follows from Eq. (21). The last step follows from simple algebra.

We prove the bound for I_1 .

We have

$$I_1 = 2d - (a + b) \leq \eta(\tau_T^+ + \tau_T^-) - 2\eta\tau_0 \leq 2\eta(\tau_0 + d) - 2\eta\tau_0 = 2\eta d. \tag{26}$$

The first step follows from Eq. (17). The second step follows by adding the two inequalities in Eq. (22) and rearranging. The third step follows from Eq. (20). The last step follows from simple algebra.

We prove the bound for I_2 .

We have

$$I_2 = 2d - (c + e) \leq \eta(\tau_W^+ + \tau_W^-) - 2\eta\tau_0 \leq 2\eta d + \eta(I_2 - I_1). \quad (27)$$

The first step follows from Eq. (17). The second step follows by adding the two inequalities in Eq. (23) and rearranging. The last step follows from Eq. (21).

Eq. (11) gives $1 - \eta > 0$. We have

$$I_2 \leq \frac{2\eta d - \eta I_1}{1 - \eta}. \quad (28)$$

The step follows from Eq. (27) by simple algebra.

We now enter the endgame. Define the three averaged Ruzsa-distance terms by

$$D_1 := d[(T; V_0) \mid (\overline{W}, S)], \quad D_2 := d[(V_0; W) \mid (\overline{T}, S)], \quad D_3 := d[(W; T) \mid (\overline{V}, S)]. \quad (29)$$

The terms D_1 , D_2 , and D_3 use the jointly conditioned Ruzsa distance in Definition 3.3. Liao records the following entropic Balog–Szemerédi–Gowers estimate.

$$D_1 + D_2 + D_3 \leq 3I_1 + 6I_2. \quad (30)$$

The step follows from Liao [Lia24, Eq. (4)]. Liao derives this estimate from the entropic Balog–Szemerédi–Gowers lemma of Gowers, Green, Manners, and Tao [GGMT25, Lemma A.2]. Define the corresponding averaged penalty totals.

$$\begin{aligned} E_1 &:= \tau_A(T \mid \overline{W}, S) + \tau_B(V_0 \mid \overline{W}, S), \\ E_2 &:= \tau_A(V_0 \mid \overline{T}, S) + \tau_B(W \mid \overline{T}, S), \\ E_3 &:= \tau_A(W \mid \overline{V}, S) + \tau_B(T \mid \overline{V}, S). \end{aligned} \quad (31)$$

For every admissible penalty τ , we have

$$\begin{aligned} \tau(T \mid \overline{W}, S) &= \tau(T + \overline{W} \mid \overline{W}, S) \\ &= \tau(V_0 \mid \overline{W}, S), \tau(T \mid \overline{V}, S) = \tau(V_0 \mid \overline{T}, S), \tau(W \mid \overline{T}, S) = \tau(W \mid \overline{V}, S). \end{aligned} \quad (32)$$

The first step follows from translation invariance because $\overline{W}$ is fixed under the conditioning. The second step follows from Eq. (14). The third equality follows by swapping the i.i.d. variables Y_1, Y_2 , which interchanges T with V_0 and $\overline{T}$ with $\overline{V}$. The last equality follows from the same swap, which fixes W and S .

We now prove the three fibre-side estimates separately. These are Liao’s inequalities [Lia24, Eqs. (15)–(17)], but we include the argument because using the three fibre inequalities without their sum-side counterparts is the new step in our proof.

We have

$$\begin{aligned} E_1 &= \tau_A(T \mid \overline{W}, S) + \tau_B(V_0 \mid \overline{W}, S) \\ &= \tau_A(T \mid \overline{W}, S) + \tau_B(T \mid \overline{W}, S) \\ &= \tau_A(T \mid W, \overline{W}) + \tau_B(T \mid W, \overline{W}) \end{aligned}$$

$$\begin{aligned}
&= \tau_A(X_1 + Y_1 \mid W, \overline{W}) + \tau_B(X_1 + Y_1 \mid W, \overline{W}) \\
&\leq \tau_A(X_1 \mid W) + \tau_B(Y_1 \mid \overline{W}) + H[T \mid W, \overline{W}] - \frac{1}{2}H[X_1 \mid W] - \frac{1}{2}H[Y_1 \mid \overline{W}] \\
&= \tau_W^- + H[T \mid W, \overline{W}] - \frac{1}{2}H[X_1 \mid W] - \frac{1}{2}H[Y_1 \mid \overline{W}] \\
&= \tau_W^- + e.
\end{aligned} \tag{33}$$

The first step follows from Eq. (31). The second step follows from Eq. (32). The third step follows from Eq. (14). The fourth step follows from Eq. (13). The fifth step follows by averaging the sum rule in Definition 3.5 using Definition 3.4. The sixth step follows from Eq. (19). The last step follows from Eq. (16).

We have

$$\begin{aligned}
\tau_A(T \mid \overline{V}, S) + \tau_B(T \mid \overline{V}, S) &= \tau_A(T \mid V_0, \overline{V}) + \tau_B(T \mid V_0, \overline{V}) \\
&= \tau_A(X_1 + Y_1 \mid V_0, \overline{V}) + \tau_B(X_1 + Y_1 \mid V_0, \overline{V}) \\
&\leq \tau_A(X_1 \mid V_0) + \tau_B(Y_1 \mid \overline{V}) + d[X_1 \mid V_0; Y_1 \mid \overline{V}] \\
&= \tau_A(X_1 \mid T) + \tau_B(Y_2 \mid \overline{T}) + d[X_1 \mid T; Y_2 \mid \overline{T}] \\
&= \tau_T^- + d[X_1 \mid T; Y_2 \mid \overline{T}] \\
&= \tau_T^- + b.
\end{aligned} \tag{34}$$

The first step follows from Eq. (14). The second step follows from Eq. (13). The third step follows by averaging the sum rule in Definition 3.5 using Definition 3.4. The fourth step follows by exchanging the i.i.d. variables Y_1, Y_2 . The fifth step follows from Eq. (19). The last step follows from Eq. (16).

The conditional law of (X_1, X_2) factors:

$$\begin{aligned}
\Pr[X_1 = x_1, X_2 = x_2 \mid T = t, \overline{T} = \overline{t}] &= \Pr[X_1 = x_1, X_2 = x_2 \mid X_1 + Y_1 = t, X_2 + Y_2 = \overline{t}] \\
&= \Pr[X_1 = x_1 \mid X_1 + Y_1 = t] \Pr[X_2 = x_2 \mid X_2 + Y_2 = \overline{t}] \\
&= \Pr[X_1 = x_1 \mid T = t] \Pr[X_2 = x_2 \mid \overline{T} = \overline{t}].
\end{aligned} \tag{35}$$

The first step follows from Eq. (13). The second step follows from the independence of (X_1, Y_1) and (X_2, Y_2) . The last step follows from Eq. (13).

Similarly, the conditional law of (Y_1, Y_2) factors:

$$\begin{aligned}
\Pr[Y_1 = y_1, Y_2 = y_2 \mid T = t, \overline{T} = \overline{t}] &= \Pr[Y_1 = y_1, Y_2 = y_2 \mid X_1 + Y_1 = t, X_2 + Y_2 = \overline{t}] \\
&= \Pr[Y_1 = y_1 \mid X_1 + Y_1 = t] \Pr[Y_2 = y_2 \mid X_2 + Y_2 = \overline{t}] \\
&= \Pr[Y_1 = y_1 \mid T = t] \Pr[Y_2 = y_2 \mid \overline{T} = \overline{t}].
\end{aligned} \tag{36}$$

The first step follows from Eq. (13). The second step follows from the independence of (X_1, Y_1) and (X_2, Y_2) . The last step follows from Eq. (13).

We first estimate the two penalties separately.

We have

$$\begin{aligned}
\tau_A(W \mid \overline{T}, S) &= \tau_A(W \mid T, \overline{T}) \\
&= \tau_A(X_1 + X_2 \mid T, \overline{T}) \\
&\leq \tau_A(X_1 \mid T) + \frac{1}{2}(H[W \mid T, \overline{T}] - H[X_1 \mid T]).
\end{aligned} \tag{37}$$

The first step follows from Eq. (14). The second step follows from Eq. (13). The last step follows from the sum rule in Definition 3.5, using Eq. (35).

We have

$$\begin{aligned}
\tau_B(W \mid \bar{T}, S) &= \tau_B(W + S \mid \bar{T}, S) \\
&= \tau_B(\bar{W} \mid \bar{T}, S) \\
&= \tau_B(\bar{W} \mid T, \bar{T}) \\
&= \tau_B(Y_1 + Y_2 \mid T, \bar{T}) \\
&\leq \tau_B(Y_2 \mid \bar{T}) + \frac{1}{2}(\mathbb{H}[\bar{W} \mid T, \bar{T}] - \mathbb{H}[Y_2 \mid \bar{T}]).
\end{aligned} \tag{38}$$

The first step follows from translation invariance because S is fixed under the conditioning. The second step follows from Eq. (14). The third step follows from Eq. (14). The fourth step follows from Eq. (13). The last step follows from the sum rule in Definition 3.5, using Eq. (36).

We have

$$\begin{aligned}
\tau_A(W \mid \bar{T}, S) + \tau_B(W \mid \bar{T}, S) &\leq \tau_A(X_1 \mid T) + \frac{1}{2}(\mathbb{H}[W \mid T, \bar{T}] - \mathbb{H}[X_1 \mid T]) + \tau_B(W \mid \bar{T}, S) \\
&\leq \tau_A(X_1 \mid T) + \tau_B(Y_2 \mid \bar{T}) + \frac{1}{2}(\mathbb{H}[W \mid T, \bar{T}] - \mathbb{H}[X_1 \mid T]) \\
&\quad + \frac{1}{2}(\mathbb{H}[\bar{W} \mid T, \bar{T}] - \mathbb{H}[Y_2 \mid \bar{T}]) \\
&= \tau_T^- + \frac{1}{2}(\mathbb{H}[W \mid T, \bar{T}] - \mathbb{H}[X_1 \mid T]) \\
&\quad + \frac{1}{2}(\mathbb{H}[\bar{W} \mid T, \bar{T}] - \mathbb{H}[Y_2 \mid \bar{T}]) \\
&= \tau_T^- + \mathbb{H}[V_0 \mid T, \bar{T}] - \frac{1}{2}\mathbb{H}[X_1 \mid T] - \frac{1}{2}\mathbb{H}[Y_2 \mid \bar{T}] \\
&= \tau_T^- + d[X_1 \mid T; Y_2 \mid \bar{T}] \\
&= \tau_T^- + b.
\end{aligned} \tag{39}$$

The first step follows from Eq. (37). The second step follows from Eq. (38). The third step follows from Eq. (19). The fourth step follows from Eq. (14). The fifth step follows from Definition 3.4. The last step follows from Eq. (16).

We have

$$\begin{aligned}
E_1 + E_2 + E_3 &= E_1 + \tau_A(V_0 \mid \bar{T}, S) + \tau_B(W \mid \bar{T}, S) + \tau_A(W \mid \bar{V}, S) + \tau_B(T \mid \bar{V}, S) \\
&= E_1 + \tau_A(T \mid \bar{V}, S) + \tau_B(T \mid \bar{V}, S) + \tau_A(W \mid \bar{T}, S) + \tau_B(W \mid \bar{T}, S) \\
&\leq (\tau_W^- + e) + \tau_A(T \mid \bar{V}, S) + \tau_B(T \mid \bar{V}, S) + \tau_A(W \mid \bar{T}, S) + \tau_B(W \mid \bar{T}, S) \\
&\leq (\tau_W^- + e) + (\tau_T^- + b) + \tau_A(W \mid \bar{T}, S) + \tau_B(W \mid \bar{T}, S) \\
&\leq (\tau_W^- + e) + 2(\tau_T^- + b) \\
&= \tau_W^- + e + 2(\tau_T^- + b).
\end{aligned} \tag{40}$$

The first step follows from Eq. (31). The second step follows from Eq. (32). The third step follows from Eq. (33). The fourth step follows from Eq. (34). The fifth step follows from Eq. (39). The last step follows from simple algebra.

This is the point where we depart from Liao's averaging argument.

Set $\tau_{\text{eg}} := E_1 + E_2 + E_3$.

We have

$$\begin{aligned}
\tau_{\text{eg}} &= E_1 + E_2 + E_3 \\
&\leq \tau_W^- + e + 2(\tau_T^- + b) \\
&= \tau_W^- + 2\tau_T^- + e + 2b \\
&\leq \tau_0 + d + \frac{1}{2}(I_2 - I_1) + 2\tau_T^- + e + 2b \\
&\leq 3\tau_0 + 3d + \frac{1}{2}(I_2 - I_1) + e + 2b \\
&\leq 3\tau_0 + 3d + \frac{1}{2}(I_2 - I_1) + (1 + \eta)d - I_2 + \frac{\eta}{2}(I_2 - I_1) + 2b \\
&\leq 3\tau_0 + 3d + \frac{1}{2}(I_2 - I_1) + (1 + \eta)d - I_2 + \frac{\eta}{2}(I_2 - I_1) + 2((1 + \eta)d - I_1) \\
&= 3\tau_0 + (6 + 3\eta)d - \frac{5 + \eta}{2}I_1 + \frac{\eta - 1}{2}I_2.
\end{aligned} \tag{41}$$

The first step follows from the definition of τ_{eg} . The second step follows from Eq. (40). The third step follows from simple algebra. The fourth step follows from Eq. (21). The fifth step follows from Eq. (20). The sixth step follows from Eq. (25). The seventh step follows from Eq. (24). The last step follows from simple algebra.

For $i \in \{1, 2, 3\}$, set $\phi_i := D_i + \eta E_i$. Then ϕ_1 , ϕ_2 , and ϕ_3 are the averaged potentials of the endgame candidates in Eq. (29), respectively.

Set

$$A_\eta := 3 - \frac{\eta(5 + \eta)}{2}, \quad B_\eta := 6 + \frac{\eta(\eta - 1)}{2}. \tag{42}$$

We have

$$\begin{aligned}
\phi_1 + \phi_2 + \phi_3 &= D_1 + D_2 + D_3 + \eta(E_1 + E_2 + E_3) \\
&= D_1 + D_2 + D_3 + \eta\tau_{\text{eg}} \\
&\leq 3I_1 + 6I_2 + \eta\tau_{\text{eg}} \\
&\leq 3I_1 + 6I_2 + \eta(3\tau_0 + (6 + 3\eta)d - \frac{5 + \eta}{2}I_1 + \frac{\eta - 1}{2}I_2) \\
&= 3\eta\tau_0 + \eta(6 + 3\eta)d + \left(3 - \frac{\eta(5 + \eta)}{2}\right)I_1 + \left(6 + \frac{\eta(\eta - 1)}{2}\right)I_2 \\
&= 3\eta\tau_0 + \eta(6 + 3\eta)d + A_\eta I_1 + B_\eta I_2.
\end{aligned} \tag{43}$$

The first step follows from the definitions of ϕ_1 , ϕ_2 , and ϕ_3 . The second step follows from the definition of τ_{eg} . The third step follows from Eq. (30). The fourth step follows from Eq. (41). The fifth step follows from simple algebra. The last step follows from Eq. (42).

We have

$$u := 2\eta d - I_1 \geq 0. \tag{44}$$

The inequality follows from Eq. (26).

We have

$$B_\eta = 6 + \frac{\eta(\eta - 1)}{2} = 6 - \frac{\eta(1 - \eta)}{2} \geq 6 - \frac{1}{8} > 0. \tag{45}$$

The first step follows from Eq. (42). The second step follows from simple algebra. The third step follows from the elementary quadratic bound $\eta(1 - \eta) \leq 1/4$. The last step follows from simple algebra.

We have

$$-6 + 23\eta - 5\eta^2 < -6 + 23\eta < -6 + \frac{23}{7} = -\frac{19}{7} < 0. \quad (46)$$

The first step follows from Eq. (11). The second step follows from Eq. (11). The third step follows from simple algebra. The last step follows from simple algebra.

We have

$$2(1 - \eta) > 0. \quad (47)$$

The step follows from Eq. (11).

Eq. (47) shows that division by $2(1 - \eta)$ preserves strict inequalities. We have

$$\frac{-6 + 23\eta - 5\eta^2}{2(1 - \eta)} < 0. \quad (48)$$

The step follows from Eq. (46).

Eq. (44) gives $u \geq 0$. We have

$$\frac{-6 + 23\eta - 5\eta^2}{2(1 - \eta)}u \leq 0. \quad (49)$$

The step follows from Eq. (48).

We have

$$\begin{aligned} \phi_1 + \phi_2 + \phi_3 &\leq 3\eta\tau_0 + \eta(6 + 3\eta)d + A_\eta I_1 + B_\eta I_2 \\ &= 3\eta\tau_0 + \eta(6 + 3\eta)d + A_\eta(2\eta d - u) + B_\eta I_2 \\ &\leq 3\eta\tau_0 + \eta(6 + 3\eta)d + A_\eta(2\eta d - u) + B_\eta \frac{2\eta d - \eta I_1}{1 - \eta} \\ &= 3\eta\tau_0 + \eta(6 + 3\eta)d + A_\eta(2\eta d - u) + B_\eta \frac{2\eta d - \eta(2\eta d - u)}{1 - \eta} \\ &= 3\eta\tau_0 + \eta(6 + 3\eta)d + A_\eta(2\eta d - u) + B_\eta(2\eta d + \frac{\eta}{1 - \eta}u) \\ &= 3\eta\tau_0 + (24\eta - 3\eta^2)d + \frac{-6 + 23\eta - 5\eta^2}{2(1 - \eta)}u \\ &\leq 3\eta\tau_0 + (24\eta - 3\eta^2)d. \end{aligned} \quad (50)$$

The first step follows from Eq. (43). The second step follows from Eq. (44). Eq. (45) shows that multiplication by B_η preserves the inequality in Eq. (28). The third step follows from Eq. (28). The fourth step follows from Eq. (44). The fifth step follows from simple algebra. The sixth step follows from simple algebra. The last step follows from Eq. (49).

We have

$$3 - (24\eta - 3\eta^2) = 3(4 - \sqrt{15} - \eta)(4 + \sqrt{15} - \eta) > 0. \quad (51)$$

The first step follows from simple algebra. The last step follows from Eq. (11).

Since $d > 0$, multiplication by d preserves strict inequalities. We have

$$\begin{aligned} \phi_1 + \phi_2 + \phi_3 &\leq 3\eta\tau_0 + (24\eta - 3\eta^2)d \\ &< 3d + 3\eta\tau_0 \\ &= 3(d[X; Y] + \eta\tau_0) \\ &= 3(d[X; Y] + \eta(\tau_A(X) + \tau_B(Y))) \\ &= 3\phi[X; Y]. \end{aligned} \quad (52)$$

The first step follows from Eq. (50). The second step follows from Eq. (51). The third step follows

from the definition of d in Eq. (16). The fourth step follows from the definition of τ_0 in Eq. (16). The last step follows from Eq. (12).

By Eq. (52), one endgame candidate has smaller averaged potential. Fixing its conditioning values gives G -valued random variables X' and Y' such that

$$\phi[X'; Y'] < \phi[X; Y]. \quad (53)$$

The step follows by fixing the conditioning values in Eq. (52).

The space of pairs of distributions on finite G is compact. Thus ϕ has a minimizer (X^*, Y^*) . If $d[X^*; Y^*] > 0$, Eq. (53) produces a pair with smaller potential, contradicting the minimality of (X^*, Y^*) . We have

$$d[X^*; Y^*] = 0. \quad (54)$$

The step follows from the contradiction argument above. By the zero-distance characterization [GGMT25, Lemma 2.2], which uses Tao's equality case [Tao10, Theorem 1.11(i)], there is a subspace $V_\eta \leq G$ such that X^* and Y^* are uniform on translates of V_η .

Eq. (11) gives $\eta > 0$, so division by η preserves inequalities. We have

$$\begin{aligned} \tau_A(U_{V_\eta}) + \tau_B(U_{V_\eta}) &= \tau_A(X^*) + \tau_B(Y^*) \\ &= \tau_A(X^*) + \tau_B(Y^*) + \frac{1}{\eta}d[X^*; Y^*] \\ &= \frac{1}{\eta}\phi[X^*; Y^*] \\ &\leq \frac{1}{\eta}\phi[X; Y] \\ &= \tau_A(X) + \tau_B(Y) + \frac{1}{\eta}d[X; Y]. \end{aligned} \quad (55)$$

The first step follows from translation invariance of the penalties. The second step follows from Eq. (54). The third step follows from Eq. (12). The fourth step follows from the minimality of (X^*, Y^*) . The last step follows from Eq. (12).

Eq. (55) holds for every $0 < \eta < 4 - \sqrt{15}$. Since G has finitely many subspaces, set

$$m := \min_{H \leq G} (\tau_A(U_H) + \tau_B(U_H)). \quad (56)$$

We have

$$m \leq \tau_A(U_{V_\eta}) + \tau_B(U_{V_\eta}) \leq \tau_A(X) + \tau_B(Y) + \frac{1}{\eta}d[X; Y]. \quad (57)$$

The first step follows from Eq. (56). The last step follows from Eq. (55).

Eq. (57) holds for every $0 < \eta < 4 - \sqrt{15}$. Choose a subspace $V \leq G$ attaining m .

We have

$$\begin{aligned} \tau_A(U_V) + \tau_B(U_V) &= m \\ &\leq \tau_A(X) + \tau_B(Y) + \frac{1}{4 - \sqrt{15}}d[X; Y] \\ &= \tau_A(X) + \tau_B(Y) + (4 + \sqrt{15})d[X; Y]. \end{aligned}$$

The first step follows from the choice of V . The second step follows by letting $\eta \uparrow 4 - \sqrt{15}$ in Eq. (57). The last step follows from simple algebra.

This proves the lemma. $\square$

The following result sharpens Liao's entropic coefficient 10 [Lia24, Theorem 4] to $6 + \sqrt{15}$.

Corollary 4.2 (Improved entropic exponent). *Suppose X and Y are G -valued random variables. Then there exists a subspace $V \leq G$ such that*

$$d[X; U_V] + d[Y; U_V] \leq (6 + \sqrt{15})d[X; Y].$$

Proof. We use Lemma 4.1 with $\tau_A(Z) := d[Y; Z]$ and $\tau_B(Z) := d[X; Z]$. We verify admissibility. For a fixed random variable R , set $\sigma_R(Z) := d[R; Z]$, using an independent copy of R whenever needed. Suppose Z_1, Z_2 are independent.

We have

$$\begin{aligned} \sigma_R(Z_1 + Z_2) &= H[R - Z_1 - Z_2] - \frac{1}{2}H[R] - \frac{1}{2}H[Z_1 + Z_2] \\ &= \sigma_R(Z_1) + H[R - Z_1 - Z_2] - H[R - Z_1] - \frac{1}{2}(H[Z_1 + Z_2] - H[Z_1]) \\ &\leq \sigma_R(Z_1) + H[Z_1 + Z_2] - H[Z_1] - \frac{1}{2}(H[Z_1 + Z_2] - H[Z_1]) \\ &= \sigma_R(Z_1) + \frac{1}{2}(H[Z_1 + Z_2] - H[Z_1]). \end{aligned} \tag{58}$$

The first step follows from the definition of σ_R . The second step follows from simple algebra. The third step follows from the Kaimanovich–Vershik–Madiman inequality [GGMT25, Lemma A.1]. The last step follows from simple algebra.

Suppose Q is jointly distributed with Z_1 .

We have

$$\begin{aligned} \sigma_R(Z_1 | Q) - \sigma_R(Z_1) &= H[R - Z_1 | Q] - H[R - Z_1] + \frac{1}{2}(H[Z_1] - H[Z_1 | Q]) \\ &\leq \frac{1}{2}(H[Z_1] - H[Z_1 | Q]). \end{aligned} \tag{59}$$

The first step follows from the definition of σ_R . The second step follows because conditioning does not increase entropy.

Thus Eqs. (58) and (59) verify the two admissibility inequalities for σ_R . These are also the calculations used by Liao [Lia24, Proof of Theorem 4]. Continuity follows from continuity of entropy on the finite probability simplex, and translation invariance follows from Definition 3.2. Thus both penalties are admissible. Apply Lemma 4.1 to obtain a subspace $V \leq G$.

We have

$$\begin{aligned} d[X; U_V] + d[Y; U_V] &= \tau_A(U_V) + \tau_B(U_V) \\ &\leq \tau_A(X) + \tau_B(Y) + (4 + \sqrt{15})d[X; Y] \\ &= d[Y; X] + d[X; Y] + (4 + \sqrt{15})d[X; Y] \\ &= 2d[X; Y] + (4 + \sqrt{15})d[X; Y] \\ &= (6 + \sqrt{15})d[X; Y]. \end{aligned}$$

The first step follows from the definitions of τ_A and τ_B . The second step follows from Lemma 4.1. The third step follows from the definitions of τ_A and τ_B . The fourth step follows from the symmetry of Ruzsa distance. The last step follows from simple algebra. $\square$

We now apply the improved penalty lemma directly with Liao's KL penalties to obtain a covering bound.

Theorem 4.3 (Our result, formal version of Theorem 1.1). *Suppose $A \subseteq \mathbb{F}_2^n$ is nonempty. Suppose $K \geq 1$ and $|A+A| \leq K|A|$. Then A is covered by at most $2K^{5+\sqrt{15}}$ translates of a subspace $H \leq \mathbb{F}_2^n$ with $|H| \leq |A|$.*

Proof. Fix A and K as in the theorem. For a G -valued random variable X , define

$$\tau^-(X) := \inf_T D_{\text{KL}}(X \parallel U_A + T), \tau^+(X) := \tau^-(X) + H[X] - H[U_A], \tau := \frac{1}{2}(\tau^- + \tau^+). \quad (60)$$

The infimum in Eq. (60) is over G -valued random variables T , and the sum $U_A + T$ uses independent copies.

Liao proves the two admissibility inequalities and translation invariance for τ [Lia24, Lemmas 10 and 13 and the proof of Theorem 2]. It remains to verify that τ is finite and continuous. Define $\mathcal{C}_A := \{U_A + T : T \text{ is a } G\text{-valued random variable}\}$. This set is the image of the compact probability simplex on G under convolution with U_A ; hence it is compact and convex. It contains the full-support distribution U_G . For every distribution P , the function $Q \mapsto D_{\text{KL}}(P \parallel Q)$ is lower semicontinuous. It therefore attains a finite minimum on $\mathcal{C}_A$.

By the definition of $\mathcal{C}_A$, the infimum in Eq. (60) ranges precisely over $\mathcal{C}_A$. We have

$$\tau^-(P) = \min_{Q \in \mathcal{C}_A} D_{\text{KL}}(P \parallel Q). \quad (61)$$

We first prove lower semicontinuity. Let $P_k \rightarrow P$. Pass to a subsequence, still denoted by (P_k) , such that $\tau^-(P_k)$ converges to $\liminf_{k \rightarrow \infty} \tau^-(P_k)$. For every k , choose $Q_k \in \mathcal{C}_A$ attaining the minimum in Eq. (61). By compactness, a further subsequence satisfies $Q_{k_j} \rightarrow Q$ for some $Q \in \mathcal{C}_A$.

We have

$$\tau^-(P) \leq D_{\text{KL}}(P \parallel Q) \leq \liminf_{j \rightarrow \infty} D_{\text{KL}}(P_{k_j} \parallel Q_{k_j}) = \liminf_{j \rightarrow \infty} \tau^-(P_{k_j}) = \liminf_{k \rightarrow \infty} \tau^-(P_k). \quad (62)$$

Because $Q \in \mathcal{C}_A$, the first step follows from Eq. (61). The second step follows from joint lower semicontinuity of KL divergence. The third step follows from Eq. (61). The last step follows from the choice of the subsequence (P_{k_j}) . Thus Eq. (62) proves that τ^- is lower semicontinuous.

We now prove upper semicontinuity. Choose $Q \in \mathcal{C}_A$ attaining $\tau^-(P)$. The minimum is finite, so P is absolutely continuous with respect to Q . For $0 < \epsilon < 1$, set $Q_\epsilon := (1 - \epsilon)Q + \epsilon U_G$. The convexity of $\mathcal{C}_A$ and the inclusions $Q, U_G \in \mathcal{C}_A$ give $Q_\epsilon \in \mathcal{C}_A$. Moreover, Q_ϵ has full support.

Set $L := \limsup_{k \rightarrow \infty} \tau^-(P_k)$. Because $Q_\epsilon \in \mathcal{C}_A$, Eq. (61) gives

$$L \leq \lim_{k \rightarrow \infty} D_{\text{KL}}(P_k \parallel Q_\epsilon), \quad 0 < \epsilon < 1. \quad (63)$$

We have

$$\limsup_{k \rightarrow \infty} \tau^-(P_k) = L \leq \lim_{\epsilon \downarrow 0} \lim_{k \rightarrow \infty} D_{\text{KL}}(P_k \parallel Q_\epsilon) = \lim_{\epsilon \downarrow 0} D_{\text{KL}}(P \parallel Q_\epsilon) = D_{\text{KL}}(P \parallel Q) = \tau^-(P). \quad (64)$$

The first step follows from the definition of L . The second step follows by letting $\epsilon \downarrow 0$ in Eq. (63). The third step follows from continuity of KL divergence in its first argument because Q_ϵ has full support. The fourth step follows because the alphabet is finite and P is absolutely continuous with respect to Q . The last step follows from Eq. (61). Thus Eq. (64) proves that τ^- is upper semicontinuous. Hence τ^- is continuous. Entropy is continuous on a finite alphabet, so τ^+ and τ are continuous as well. Thus τ satisfies every hypothesis of Definition 3.5.

By the nonnegativity of KL divergence, $\tau^-(U_A) \geq 0$. Taking $T \equiv 0$ in the definition of τ^- gives $\tau^-(U_A) \leq D_{\text{KL}}(U_A \parallel U_A) = 0$. We have

$$\tau^-(U_A) = 0. \quad (65)$$

We have

$$\tau^+(U_A) = \tau^-(U_A) + H[U_A] - H[U_A] = 0 + H[U_A] - H[U_A] = 0. \quad (66)$$

The first step follows from the definition of τ^+ in Eq. (60). The second step follows from Eq. (65). The last step follows from simple algebra.

We have

$$\tau(U_A) = \frac{1}{2}(\tau^-(U_A) + \tau^+(U_A)) = \frac{1}{2}(0 + \tau^+(U_A)) = \frac{1}{2}(0 + 0) = 0. \quad (67)$$

The first step follows from the definition of τ in Eq. (60). The second step follows from Eq. (65). The third step follows from Eq. (66). The last step follows from simple algebra.

Apply Lemma 4.1 with $X = Y = U_A$ and $\tau_A = \tau_B = \tau$. The lemma gives a subspace $V \leq G$.

We have

$$\begin{aligned} d[U_A; U_A] &= H[U_A - U_A] - H[U_A] \\ &= H[U_A + U_A] - H[U_A] \\ &\leq \log |A + A| - H[U_A] \\ &= \log |A + A| - \log |A| \\ &\leq \log(K|A|) - \log |A| \\ &= \log K. \end{aligned} \quad (68)$$

The first step follows from Definition 3.2. The second step follows because subtraction and addition agree in $\mathbb{F}_2^n$. The third step follows from the entropy bound by the logarithm of the support size. The fourth step follows from the uniformity of U_A . The fifth step follows from the hypothesis $|A + A| \leq K|A|$. The last step follows from simple algebra.

We have

$$\begin{aligned} \tau^-(U_V) + \tau^+(U_V) &= 2\tau(U_V) \\ &\leq 2\tau(U_A) + (4 + \sqrt{15})d[U_A; U_A] \\ &= (4 + \sqrt{15})d[U_A; U_A] \\ &\leq (4 + \sqrt{15})\log K. \end{aligned} \quad (69)$$

The first step follows from Eq. (60). The second step follows from Lemma 4.1. The third step follows from Eq. (67). The last step follows from Eq. (68).

There exists an element $t \in G$ such that

$$|A \cap (V + t)| \geq 2^{-(4+\sqrt{15})\log K} \max\{|A|, |V|\} = K^{-(4+\sqrt{15})} \max\{|A|, |V|\}. \quad (70)$$

The first step follows from Liao's dense-coset consequence [Lia24, Claim 12], applied to Eq. (69). The last step follows from simple algebra. Liao derives the dense-coset consequence from the exact slice identities in [Lia24, Lemma 11].

Set $R := K^{4+\sqrt{15}}$. Since $K \geq 1$, we have $R \geq 1$.

We have

$$|A \cap (V + t)| \geq K^{-(4+\sqrt{15})} \max\{|A|, |V|\} = \frac{\max\{|A|, |V|\}}{R}. \quad (71)$$

The first step follows from Eq. (70). The last step follows from the definition of R .

Apply Lemma 3.6 using Eq. (71) and $R \geq 1$. The set A is covered by at most $2KR$ translates of a subspace $H \leq G$ with $|H| \leq |A|$.

We have

$$2KR = 2K^{1+4+\sqrt{15}} = 2K^{5+\sqrt{15}}.$$

The first step follows from the definition of R . The last step follows from simple algebra. Thus A is covered by at most $2K^{5+\sqrt{15}}$ translates of H . $\square$

GenAI Usage Disclosure

Codex were used to polish language, such as grammar and wording.

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